

# IVAN ARJUNE

347.781.4826 | [iarjune@gmail.com](mailto:iarjune@gmail.com) | NYC | <https://iarjune.github.io/portfolio/>

## LEAD INFRASTRUCTURE & MLOPS ENGINEER

Production machine learning is only as effective as the infrastructure, data pipelines, and operational systems supporting it. Moving models from experimentation to production—and keeping them reliable, performant, and relevant over time—requires more than ML tooling. It requires a deep understanding of distributed systems, data engineering, infrastructure automation, and production operations.

I bring this systems-first perspective to MLOps, combining extensive experience in cloud-native architecture, Kubernetes orchestration, streaming data platforms, and GitOps to build the foundations that make machine learning operationally sustainable. I've engineered high-throughput data infrastructure and low-latency retrieval systems, supporting real-time workloads where performance, availability, and predictable scaling directly affect business outcomes.

My focus is on eliminating the operational friction between model development and production. By integrating Argo Workflows, Argo Events, Kubeflow, MLflow, and feature stores, I've built automated DataOps and MLOps workflows that connect event-driven data processing, model retraining, and deployment. These Kubernetes-native architectures replaced legacy Hadoop and Airflow-based workflows with more automated, reproducible, and operationally consistent pipelines.

Beyond orchestration, I engineer the production environment that ML systems depend on: scalable compute, reliable streaming, observability, security guardrails, and cost-efficient resource utilization. My experience operating complex systems like Kafka, Aerospike, ClickHouse, and Vespa bridges the gap between data-intensive ML workloads and the demands of real-time production systems.

I build MLOps platforms that turn machine learning from a collection of experimental workflows into a repeatable, scalable, and production-ready engineering discipline

## SKILLS

- **Data Engineering & Distributed Systems:** Apache Kafka, Kafka Connect, MirrorMaker, Schema Registry, RabbitMQ, Aerospike, ClickHouse, Redis, Vespa Cloud, Vector Search, Real-Time Streaming, Distributed Storage, Low-Latency Data Retrieval
- **MLOps, DataOps & ML Platforms:** Kubeflow, MLflow, Argo Workflows, Argo Events, Feature Stores, Automated Model Retraining, Event-Driven Pipelines, Metric Drift Detection, Apache Spark, Hadoop, Airflow
- **CI/CD & GitOps:** Argo CD, Argo Rollouts, Argo Workflows, GitHub Actions, Jenkins, Bitbucket Pipelines, Kustomize, GitOps, Progressive Delivery, Automated Integration Testing
- **SRE, Observability & Reliability:** Datadog, OpenTelemetry, Prometheus, Grafana, PagerDuty, Distributed Tracing, Incident Response, Performance Optimization, High Availability, Reliability Engineering, MTTR Reduction
- **DevSecOps & AI Infrastructure:** Open Policy Agent (OPA), Snyk, JFrog Xray, SonarQube, SBOM, Secrets Management, IAM, HashiCorp Vault, Akeyless, LLM Gateways, PII Sanitization, AI-Assisted Engineering
- **Cloud & Infrastructure:** AWS (EKS, MSK), Hybrid Cloud Architecture, Multi-Account AWS, Kubernetes, Docker, KVM, Rancher, Crossplane
- **Infrastructure as Code & Automation:** Terraform, Puppet, Ansible, Python, Bash

## EXPERIENCE

adMarketplace | New York, NY Staff Platform & Infrastructure Engineer | Oct 2016 – May 2026

- **Architected multi-account, multi-region AWS cloud infrastructure** centered on Amazon EKS across Dev, Staging, Production, and Management environments using Terraform, ArgoCD, and Crossplane.
- **Reduced enterprise AWS cluster spend by 50% (\$600K+ in annualized OpEx savings)** by implementing Karpenter dynamic autoscaling, aggressive spot instance strategies, Kubecost cost-allocation, and automated right-sizing algorithms.
- **Modernized core infrastructure and data workloads** by replacing legacy batch frameworks (Airflow, Hadoop) with Kubernetes-native Argo Workflows, ArgoCD, and Kubeflow platforms using declarative GitOps.

- **Replaced fragmented legacy monitoring** with an enterprise-grade Datadog APM, Tracing and PagerDuty observability stack, accelerating incident response and root-cause analysis across core microservices and ML pipelines.
- **Strengthened infrastructure security and audit posture** by embedding Snyk, SonarQube, and JFrog Artifactory into CI/CD pipelines, enforcing SOC 2 compliance and automated SBOM generation.
- **Managed high-throughput Amazon MSK (Apache Kafka) data streaming clusters**, processing 2B+ daily events (500+ GB/day) for real-time distributed ad serving, Feature Stores, and downstream ML prediction engines.
- **Architected and operated production Vespa Cloud search infrastructure**, leveraging custom schema configurations and multi-phase tensor ranking to achieve sub-20ms latency at 15,000+ peak QPS across hybrid vector and BM25 indices.
- **Provisioned and optimized high-speed distributed storage systems** (ClickHouse, Aerospike, Redis) for real-time feature retrieval, distributed ad processing, and low-latency system evaluation.
- **Orchestrated Kubeflow continuous retraining pipelines and MLflow model registries** using Python and Bash automation, processing 50M+ vector embeddings daily on Kubernetes.
- **Built and maintained centralized LLM proxy gateways** with automated PII sanitization scripts, enabling safe enterprise adoption of AI coding tools (Claude Code, GitHub Copilot) across engineering teams.
- Built declarative Infrastructure as Code (IaC) pipelines using Terraform, ArgoCD, and Crossplane to automate complex distributed cloud topologies.

## EDUCATION & CERTIFICATIONS

- **New York Institute of Technology** — B.S. in Computer Science
- **Certification:** AWS Certified Cloud Practitioner